

spec

Feature Specification: Knowledge Areas & Bidirectional Deep Linking

Feature Branch: 002-knowledge-areas-deep-linking

Created: 2026-09-01

Status: Draft

Input: User description:

“No knowledge covered - areas - like we have in AI Curriculum (ai_interviewing)! We have to extract all terms and areas and for them all to create same materials kinds like ai_interviewing has! All areas and terms mentioned in transcript and video MUST BE linked to these knowledge resources - areas with chapters and quizzes! Extend this properly! Use the same technology we have used to create UI / UX for ai_interviewing! Design unique look and feel using OpenDesign... Search MUST be able to search all contents - areas with chapters, quizzes, everything from the interview and to link exactly where they appear! In videos we MUST BE linked to exact times, and from whole transcripts to jump to whole areas, proper chapters and part of the chapters (or quizzes / tests) where content is mentioned and where it appears!”

Amended the same day by a second operator directive, which reframes the work rather than adding to it:

“Use ai_interviewing as reference and base on top of which we shall build our workshop! Do not clone or copy it but use as the reference / port it properly (parts we may need)! Current workshop is extremely poorly made!”

And amended again, with the two corrections that decide how the work is judged:

“Do not just pickup materials from ai_interviewing but create the materials which are following our chapters (chapter 01 we have currently), and then as we are adding more chapters after they are processed we MUST extract all areas and themes which will be fully covered through added areas and chapters with exhaustive long and short tests / quizzes! Everything MUST BE cross-referenced, searchable by meaning and context and fully connected! Transcripts and video content, codebases, knowledge base - areas with chapters, quizzes / tests - all of it!”

Why this feature exists: the workshop is the wrong kind of artefact

This framing is stated first because it is the reason the feature is not an add-on.

Measured 2026-09-01, comparing the reference module with what feature 001 produced:

Dimension	ai_interviewing (reference)	workshop (today)
Frontend components	8	10
Frontend lines of code	2,690	4,566
Backend API routes	4	25
Content files	68, plus a taxonomy of areas	8
Knowledge areas with materials	34 area documents	0
Questions with source provenance	0 of ~68 files	n/a — no questions exist

The workshop has **more** code and almost none of the substance. The component vocabulary shows why:

- ai_interviewing: **module · practice · progress · plans** — a *learning platform*.

- workshop: **chapter-list** · **transcript** · **recording-player** · **crossrefs**
— a *media viewer*.

The workshop is not badly built. It is an excellent document browser over a recording, when what is wanted is a curriculum a person learns from. Feature 001 delivered the substrate — a faithful transcript, 1,101 addressable passages, a working search and a cross-reference graph. Feature 002 delivers the missing half: the knowledge layer that turns that substrate into a course.

And it does one thing the reference cannot do at all. The reference's questions carry a prompt and an answer and nothing else — no link to any source. The workshop's 1,101 passages each carry a minted identifier that has been *measured* to survive both a content correction and a structural insertion. So every workshop question can cite the exact moment that teaches it, and a learner can jump from a quiz question to that moment in the recording and back. That is the game-changer, and it is the operator's "*linked to exact times ... jump to whole areas, proper chapters and part of the chapters (or quizzes / tests)*" stated as a mechanism.

Context: Verified Starting State

Measured on 2026-09-01. Every figure below is re-derivable by the command in the right-hand column; none is quoted from a document. Requirements later in this specification are only meaningful against these values.

Fact	Measured value	Re-derive with
Platform reachable	live on the LAN, HTTP 200 on /api/ health	<code>curl -s -o /dev/null -w '%{http_code}' http:// <host>:<port>/api/health</code>
Passages in the corpus	1,101	count records in the passage registry
— transcript segments	1,055	group the registry by kind
— document sections	44	”
— code passages	2	”
Passages flagged uncertain	267 (24.2%)	<code>count uncertain == true</code>
	1	<code>count distinct scope</code>

Fact	Measured value	Re-derive with
Distinct scopes (chapters) present		
Passage identifier	minted ULID, persisted, never positional, never content-derived	registry pid field
Identifier survival, measured	1,101 / 1,101 preserved across a content correction and a section insertion	replay of the 001 experiment
Segment timing granularity	median segment 6.74 s , p95 10.78 s , max 20.22 s , mean 6.56 s over 1,055 segments spanning 6,924.10 s	derive per-segment spans from the registry
Word-level timings	present, in a per-chapter sidecar — not in the registry. 15,610 word records, each carrying a start, an end and a probability; 2,717 (17.4%) flagged below the confidence threshold	read the chapter's word sidecar; inspect counts
Indexed content kinds declared by search	transcript_segment, doc_section, code, diagram — but the corpus holds 0 diagram passages, so one declared kind is empty	/api/search response corpus.indexed_kinds; group the registry by kind
Positional match offsets in a search hit	the hit carries a highlights field typed as a list of offset pairs and it is unconditionally set to empty ; the	inspect the hit-construction path and any live /api/search response

Fact	Measured value	Re-derive with
Relevance floor	snippet is generated with empty match delimiters. No offsets are produced anywhere today uncalibrated — the envelope reports floor_calibrated: false and nothing is demoted	any /api/search response
Media reachability declared by search	<i>“transcripts and captions only”</i>	/api/search response, corpus.media_reached_via
Endpoints for areas / terms / quizzes	absent — /api/areas, /api/terms, /api/quizzes all return 404 25 PASS / 4 DEBT; the 4 are permanent negative-route assertions, driven from a route manifest	probe each path
Server behaviour suite	5 areas — as prose only. Area documents exist under the workshop’s training documentation, ids 01–05, already using an adapted form of the reference skeleton. There is no machine-readable representation: no identifiers, no schema, no database table, no endpoint	the server-unity verifier
Existing workshop taxonomy		list the training area documents; search the platform for an area type or table

Fact	Measured value	Re-derive with
Existing workshop term inventory	none. No glossary, no term table, no term field anywhere in the platform or the curriculum	search the platform and curriculum for a term entity
Existing workshop assessment model	none. Quiz styling is present in the vendored design kit; the concept is not — no data model, no endpoint, no component	search for a question or quiz entity
Extracted taxonomy (landed 2026-09-01, during drafting)	7 tracks, 37 modules, 137 terms for the one chapter; linkage for all 1,101 passages; a reverse index keyed by module, term and track	count the records in the chapter's knowledge directory
Extracted taxonomy coverage	884 of 1,055 transcript passages (83.79%) and 14,290 of 15,610 words (91.54%); every uncovered passage carries a stated reason; 211 of the 267 uncertain passages are covered	read the coverage record
Chapter sections	already modelled, and derived rather than authored: transcript sections carry a rule and a cross-check; the chapter's document sections number 46	the chapter endpoint's sections; the document section sidecar
Reference area skeleton	7 level-2 headings in a fixed order,	

Fact	Measured value	Re-derive with
	present in 34/34 area documents (one document substitutes two alternatives for one heading)	count headings across the reference area documents
Reference area corpus	34 area documents in Markdown, plus 74 diagrams rendered to 3 formats each from Mermaid source	count .md files and diagram files in the reference area directory
Reference four-format coverage	25 of 34 areas carry all four formats — 9 are Markdown only	file-existence matrix over md/html/docx/pdf
Reference question bank	312 multiple-choice + 473 short-answer = 785 authored items across 34 files each, linked to an area by a code rather than by a key	count items in the reference question banks
Reference question provenance	0 of 785. The question record has fields for prompt, choices, answer, explanation, category and difficulty — and no source field of any kind	reference question type definition and bank schema
Reference search	a substring scan over three tables with a fixed result limit, no ranking and no locus ; its hit carries a type, an area id, a title and a snippet	reference search implementation
Generative model on this host	none. The local inference host	list the local model store

Fact	Measured value	Re-derive with
Entailment path	serves two models, both embedding models an entailment-based support verifier exists in the platform and is selected when a model directory is configured; the fallback is lexical overlap with a configurable floor	inspect the answering verifier wiring
Export and OCR toolchain	all present on this host: a Markdown converter, an HTML-to-PDF renderer, an office suite, a PDF text extractor, an OCR engine and a Mermaid renderer all resolve on PATH	<code>command -v each</code>

A claim this document made and now withdraws

An earlier revision of this table stated “**Word-level timings: absent.**” That was **wrong** and is withdrawn rather than quietly corrected, because the rules governing this repository require a retracted figure to be retracted visibly.

What was actually measured: the *passage registry* carries a per-segment end time and nothing finer — that half was right. But a **per-chapter word sidecar exists** alongside it and holds 15,610 word records with individual start and end times. The error was to infer the corpus from one file. Its consequence would have been material: the specification would have committed to segment-level precision as a ceiling when word-level precision is reachable by joining the sidecar on time.

The corrected position, which the requirements below are written against: **precision is word-level where the word sidecar covers the material and segment-level otherwise**, the two are not

interchangeable, and every link must state which one it used. 17.4% of words are flagged below the confidence threshold, so word-level precision is *available*, not *uniformly trustworthy*.

Three facts that are load-bearing and easy to misread

1. **The workshop is not starting from zero, and not starting from data either.** Five areas already exist as prose, in an adapted form of the reference skeleton. What does not exist is any machine-readable form of them — no identifier, no schema, no endpoint. The work is to promote prose into a first-class entity and then extend it, not to invent a taxonomy against an empty page and not to write documents that already exist.
2. **The highlights field is a measured lie waiting to happen.** A search hit already advertises a list of match offsets, and that list is unconditionally empty. A consumer reading the schema would reasonably conclude offsets are available. This is exactly the shape of defect the governing rules call a blind instrument, and the “say *where* it appears” requirement lands directly on it.
3. **There is no generative model on this host.** Anything requiring new prose to be written at run time cannot execute here today. This is why prose authorship is an open clarification rather than an assumption. Note the contrast with the export and OCR toolchain, which is fully present — so the second clarification is a question about scope, not about capability.

Clarifications

Session 2026-09-01

Three questions were left open because each genuinely changes the size and the risk of the work and none could be settled from evidence in the tree. Everything else the description left unstated was decided, and those decisions are recorded under **Assumptions** and **Resolved Decisions** with the reasoning, per the operator’s standing instruction to take the safest, most stable, risk-free option whenever a choice is forced.

Status, 2026-09-02: two of the three are now DECIDED and ONE remains open.

Clarification	Status	Where the decision lives
prose authorship	OPEN — genuinely undecided	still bounds T031; no decision record exists
what “ <i>in videos</i> ” covers	DECIDED 2026-09-02 — spoken plus on-screen OCR, extending this spec in place	D5 below; tasks T123–T142
answering over the new material	DECIDED 2026-09-02 — build the answer-against-question verifier	T115/T116, marked [UNBUILT], not [BLOCKED]

A decided clarification is not a built one. Both decided questions leave **unbuilt code** behind them, and the count that matters for planning is stated plainly in each entry rather than folded into a single “resolved”.

- **[NEEDS CLARIFICATION: prose authorship]** Area materials in the reference are written prose — “*Why this matters*”, “*Core concepts in one page*”, model answers. Three ways to produce the workshop’s equivalent differ by an order of magnitude in scope and in risk, and the host makes the difference concrete rather than theoretical, because **it has no generative model: (a) extractive only** — an area document is assembled entirely from cited passages and fixed templated connective text, so every sentence is traceable and nothing is invented, but the prose reads as an anthology rather than a lesson; **(b) agent-authored at build time** — a human or an agent working in this repository writes the prose once, cites passages for every claim, and the platform never generates text at run time; **(c) platform-generated at run time** — the platform drafts area prose itself, which requires a generative model to be installed and brings the fabrication risk this repository has already measured. *Scope impact*: (a) needs no model and no review workflow; (b) needs a review-and-publication gate; (c) needs a model, a grounding pipeline, and a fabrication measurement before anything may be published.
- **[NEEDS CLARIFICATION: what “*in videos*” covers] DECIDED 2026-09-02 — spoken PLUS on-screen OCR, extending this specification IN PLACE.** The question is kept in full below, because the decision is only legible against what was actually asked. The description requires terms mentioned “*in transcript and video*” to be linked to “*exact times*”. The corpus contains only what was

spoken. Text that appeared **on screen** — the substance of a screen recording — is not transcribed and is not in the registry, and the one declared content kind that could hold it is measured empty (diagram, 0 passages — starting-state table above). Linking on-screen occurrences would mean optical character recognition across the whole recording, producing a new passage kind, new timings and a new accuracy obligation of its own. *Scope impact:* spoken-only is a linking feature over data that already exists; spoken-plus-on-screen is a second ingestion pipeline that must be measured before it can be trusted. **This was a question about scope, not about capability** — the OCR engine resolves on this host, so the answer was never forced by the environment.

The decision, and the three parts of it that bind work. (1) **Scope:** on-screen text is **in**, so every one of the three consequences the question named is now owed — a new passage kind, new timings, and a new accuracy obligation. (2) **Placement:** this specification is extended **in place** rather than a separate feature 003 being opened, because the new kind joins the *existing* passage registry, the *existing* two-valued precision model and the *existing* mention join; a second spec would have had to restate all three and would have created the second source of truth the reconciliation rules exist to prevent. (3) **Sequencing: specify now, build later.** The work is written down as tasks **T123–T142** and is sequenced **after this specification’s 28 already-unbuilt tasks**. Nothing in this decision is built, and no checkbox was ticked for it. Full reasoning, alternatives and what would reopen it: **D5** below.

The consequence, stated plainly because it changes when this feature can close: spec 002 cannot close until OCR accuracy is MEASURED. Not designed, not implemented, not reviewed — **measured**, per SC-031 through SC-034, on the chapter the mentions are derived from. Accuracy for automatic speech recognition in this project is a measured quantity with a recorded procedure (workshop/pipeline/CALIBRATION.md, workshop/pipeline/compare_engines.py, workshop/scripts/verify-accuracy.sh — all three present, measured 2026-09-02). Optical character recognition entering the same corpus at a *lower* evidentiary standard than the speech it sits beside would make the corpus’s overall accuracy unknown while every published figure continued to look precise. **A task that says “OCR the video” without an accuracy budget is the defect this clarification was raised to prevent, and adding the tasks without the budget would have been answering the question by ignoring it.**

- **[NEEDS CLARIFICATION: answering over the new material]**

Question answering over the workshop corpus has a **known, measured, unclosed defect**: on the unanswerable set, 3 of 33 questions were answered rather than declined, and separately, *extractive evasion* is undefended — a passage fragment that is topically related but does not answer the question passes every existing check, because every layer verifies *claim against passage* and no layer verifies *claim against question*. Extending answering across areas, lessons and quiz questions multiplies the surface on which that defect can appear. *Scope impact*: closing it means building a claim-to-question relevance verifier and measuring it — and on this host, an entailment-class model would have to be installed first; leaving it open means the feature ships with the defect stated in its limits document and answering held to retrieval over the new kinds without generating answers about them. **The one option that is not available is to describe it as solved.**

User Scenarios & Testing (*mandatory*)

User Story 1 - The workshop teaches areas, not just chapters (Priority: P1)

A learner opens the workshop and sees the knowledge it covers laid out as areas — the way the reference module lays out its taxonomy — rather than as a single recording with a transcript underneath. They pick an area, read its materials, and every claim in those materials points back to the moment in the workshop where it was actually taught.

Why this priority: this is the missing substance. Without it the platform remains a media viewer, which the starting-state table measures rather than asserts: 0 areas, 0 area materials, and `/api/areas` returning 404. Every later story navigates to, from, or across areas, so nothing else can exist first.

Independent Test: deliver only the taxonomy and the area documents. A reviewer opens the area list, picks any five areas at random, and confirms each has materials in the reference skeleton and that every substantive claim carries a citation that resolves to a real passage. Value is delivered even if no new interface, search or assessment is ever built.

Acceptance Scenarios:

1. **Given** the indexed corpus, **When** taxonomy extraction is run, **Then** a set of knowledge areas and terms is produced in which every area and every term is evidenced by at least one passage, and no area or term exists without evidence.
 2. **Given** the extracted taxonomy, **When** a reviewer inspects any area, **Then** it carries a materials document following the same section skeleton the reference module uses.
 3. **Given** an area's materials, **When** a reviewer examines any substantive claim, **Then** it carries at least one passage citation that resolves, or it is visibly marked as editorial framing rather than as workshop content.
 4. **Given** extraction is run a second time on unchanged content, **When** it completes, **Then** the taxonomy is identical and no identifier has changed.
 5. **Given** a passage that no area claims, **When** the extraction completes, **Then** that passage is explicitly classified as unattached with a stated reason, rather than silently dropped from the accounting.
 6. **Given** a term the extractor recognised with low confidence, **When** it is published, **Then** it is marked uncertain and is not presented as a confidently identified term.
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User Story 2 - Go from any point to every related point, in both directions (Priority: P2)

A learner watching the recording pauses at a moment and jumps straight to the area that moment belongs to, and from there to the lesson section and to the questions that test it. Later they come at it from the other end: they open a quiz question and jump to the exact moment in the recording that teaches its answer, then come back without losing their place.

Why this priority: this is the heart of the operator's description and the mechanism every other story depends on. It is second only because it needs areas to link to. Once it exists, search results, quizzes and the learning surface all navigate through it rather than each inventing its own notion of "where".

Independent Test: with the taxonomy present, enumerate every recorded link, traverse it forward and then backward, and confirm the reverse traversal returns the origin. No interface is required — this is testable entirely through the platform's own interfaces.

Acceptance Scenarios:

1. **Given** a transcript moment, **When** a learner asks what it belongs to, **Then** the areas, the chapter section and the questions attached to it are returned, each reachable in one step.
 2. **Given** an area, **When** a learner asks where it appears, **Then** every passage that evidences it is returned, each with the position at which it occurs.
 3. **Given** a question, **When** a learner follows its citation, **Then** playback lands inside the time span of the cited passage.
 4. **Given** any recorded link, **When** it is traversed in reverse, **Then** the origin is returned; no link is one-way.
 5. **Given** a passage whose text is corrected and the content re-indexed, **When** every link is re-traversed, **Then** all of them still resolve to the same targets.
 6. **Given** a link whose target no longer exists, **When** it is followed, **Then** it fails loudly with a stated reason, and is never quietly re-pointed at a different target.
 7. **Given** a chain of related material that forms a cycle, **When** a learner traverses it, **Then** they are not trapped; the traversal terminates.
 8. **Given** a passage that has been redacted, **When** any link to it is followed, **Then** the redaction is honoured at every layer, not merely in the transcript view.
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User Story 3 - Search everything, and be told exactly where it is (Priority: P3)

A learner searches once and gets results drawn from areas, lesson sections, quiz questions, the transcript, the written documents, the code and the diagrams — and each result says *where* it occurs: which area and chapter section contain it, and at what point within them.

Why this priority: search already works over the 001 corpus, so this story extends a proven capability rather than creating one. It is third because the new content kinds must exist before they can be indexed.

Independent Test: with the new kinds indexed, run a benchmark of meaning-based queries whose expected targets are known in advance and confirm both that the expected item is returned and that every returned hit carries a locus that resolves.

Acceptance Scenarios:

1. **Given** all content kinds are indexed, **When** a learner searches, **Then** results may come from any kind, and a kind with content but no index entries is a failure rather than an empty category.
 2. **Given** a search result, **When** it is displayed, **Then** it states the unit that contains it and the position within that unit — a time span where the source is media-backed, a text span otherwise.
 3. **Given** a learner types the first characters of an area or term name, **When** suggestions appear, **Then** areas and terms are offered alongside the existing suggestions.
 4. **Given** a learner wants to narrow results, **When** they filter by area or by kind, **Then** only matching results are returned and the filter is reflected in what is shown.
 5. **Given** the search backend is degraded or unavailable, **When** a learner searches, **Then** the system says it could not answer, and does not render the failure as “no results”.
 6. **Given** the corpus has grown with the new kinds, **When** latency is measured, **Then** it still meets the thresholds established for the existing corpus.
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User Story 4 - Practice, and see exactly where each answer comes from (Priority: P4)

A learner finishes an area and practises it: multiple-choice questions, short-answer questions and flashcards. Every question shows where its answer comes from, and a single action takes the learner to that moment in the recording and back to the question afterwards.

Why this priority: this is where the workshop exceeds its reference rather than catching up to it, so it is worth building deliberately and on top of proven linking rather than early and fragile. It depends on areas (US1) and on the link mechanism (US2).

Independent Test: enumerate every served question, resolve every citation it carries, and confirm none is unresolvable; then, for questions citing a transcript segment, follow the citation and confirm playback lands inside that segment’s time span.

Acceptance Scenarios:

1. **Given** an area, **When** a learner opens its practice set, **Then** questions of the reference module's kinds are available — multiple choice, short answer and flashcard.
 2. **Given** any served question, **When** its provenance is inspected, **Then** it carries at least one passage citation that resolves.
 3. **Given** a question whose citation cannot be resolved, **When** the practice set is assembled, **Then** that question is withheld rather than served without provenance.
 4. **Given** a question citing a transcript segment, **When** the learner follows the citation, **Then** playback lands inside that segment, and returning restores the question and the learner's progress.
 5. **Given** a model answer that was assembled rather than authored, **When** it is shown, **Then** it is marked as such, and is not presented as a verified answer.
 6. **Given** a learner completes practice, **When** they return later, **Then** their progress per area and per question set is preserved.
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User Story 5 - A learning surface, with its own visual identity (Priority: P5)

A learner uses an interface built for learning — an area view, a practice view, a progress view and a study-plan view — rather than a file browser with a video in it. It looks like the workshop's own product, not like a copy of the reference module.

Why this priority: the surface is the vehicle, not the cargo. Stories 1 to 4 are all testable through the platform's interfaces without it, and the workshop already has a working, if mis-shaped, front end. Reshaping it last means it is reshaped around behaviour that has already been proven.

Independent Test: open each new view and complete its primary task using only the keyboard, then run an accessibility audit and a design-token conformance check against the whole surface.

Acceptance Scenarios:

1. **Given** the platform is running, **When** a learner opens it, **Then** the entry point is the knowledge the workshop covers, with the recording reachable from within an area rather than being the top-level object.

2. **Given** an area, **When** a learner opens it, **Then** they can read it, practise it, see their progress in it, and reach every linked point from it.
 3. **Given** the existing chapter, transcript, recording and cross-reference views, **When** the new surface is delivered, **Then** those capabilities remain reachable, subordinated to the area they belong to rather than removed.
 4. **Given** any new view, **When** its styling is inspected, **Then** every colour, spacing and typographic value resolves through the shared design-token contract, and no brand value is written literally.
 5. **Given** any new view, **When** an accessibility audit is run, **Then** it reports no Level A or Level AA violation, and every deep link is reachable and operable by keyboard alone.
 6. **Given** a learner is midway through an area, **When** they follow a deep link out and come back, **Then** their position is restored.
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User Story 6 - Take an area away with you, in four formats (Priority: P6)

A learner exports an area and gets the same set of document formats the reference module ships: Markdown, HTML, Word and PDF — with the citations preserved.

Why this priority: last because it is a transformation of material that must exist and be correct first. Exporting wrong material in four formats is four times the damage.

Independent Test: run the export over every published area and check the format matrix is complete, then re-run it and confirm the extracted text of each regenerated document is unchanged.

Acceptance Scenarios:

1. **Given** a published area, **When** export is run, **Then** all four formats are produced.
2. **Given** the export has been run, **When** an area is exported again from unchanged source, **Then** the extracted text of each output is identical to the previous run.
3. **Given** an exported document, **When** a reader inspects a cited claim, **Then** the citation is present and identifies the passage it came from.
4. **Given** an area that has not passed its publication review, **When** export is run, **Then** that area is not exported and the omission is reported.

5. **Given** a diagram belonging to an area, **When** the area is exported, **Then** the diagram is rendered into each format rather than being dropped.
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User Story 7 - Add the next chapter and get the whole knowledge layer (Priority: P7)

A maintainer finishes recording Chapter 2. They run one documented procedure. The chapter arrives transcribed, its passages minted, its areas extracted and merged into the existing taxonomy, its materials written, its short and long question sets built, everything cross-referenced, indexed and deep-linked — with no hand-assembly of structure and no code change.

Why this priority: last, because it validates every earlier story at once. It is not the least important — it is arguably the point of the whole feature, since a curriculum that cannot grow is a document. But it can only be *tested* once the things it must produce are known to be producible, so it is sequenced where it can actually be proven rather than where it feels most important.

Independent Test: run the documented procedure against a small synthetic chapter and confirm that every output appears and that the diff contains no hand-created structural file.

Acceptance Scenarios:

1. **Given** a new chapter's raw material placed where the procedure specifies, **When** the procedure runs, **Then** the chapter emerges transcribed, with minted passage identifiers, extracted areas, authored materials, short and long question sets, cross-references, index entries and deep links.
2. **Given** an existing taxonomy, **When** a new chapter is processed, **Then** established areas keep their identifiers and gain evidence, newly evidenced areas are added, and a contradiction between new evidence and an existing area is reported rather than silently resolved.
3. **Given** the procedure is run twice on the same chapter, **When** the second run completes, **Then** nothing has changed and nothing is duplicated.
4. **Given** a chapter whose material is incomplete, **When** the procedure runs, **Then** it names exactly what is missing and publishes nothing.

5. **Given** two chapters are present, **When** a learner searches or browses an area, **Then** results and evidence span both chapters, and no area is confined to the chapter it first appeared in.
 6. **Given** the procedure exists, **When** a maintainer looks for how to add a chapter, **Then** there is exactly one documented way to do it.
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Edge Cases

- **A term appears in a passage that has been redacted.** The term keeps its evidence count only if another passage supports it; otherwise the term itself must disappear from the published taxonomy, not merely lose one link. Redaction that stops at the display layer is the failure mode this repository has already contracted against.
- **A term is a common word.** “Test”, “run”, “build” occur constantly. An extractor that treats frequency as significance produces a taxonomy of stopwords. Significance must be demonstrated, and the demonstration must be recorded so it can be challenged.
- **An area is evidenced only by passages flagged uncertain.** 267 of 1,101 passages are uncertain. An area resting entirely on uncertain evidence must be marked as such and must not read like a confident one.
- **A mention spans a segment boundary.** A term spoken across the seam between two segments belongs to both; attaching it to one silently loses half its evidence.
- **Two areas overlap almost completely.** Near-duplicate areas fragment the taxonomy and split their own evidence. Merging is a decision with consequences for every link already pointing at either one.
- **A question’s only citation is corrected after the question is written.** The citation must survive the correction; that is exactly what the minted identifier exists for, and it must be proven, not assumed.
- **A question cites a passage in a different chapter than the area it belongs to.** Legitimate once more than one chapter exists; the model must not assume an area lives in one chapter.
- **The export toolchain is not installed.** Four-format export depends on document conversion tooling. Its absence must report that the export could not be performed — never that it was performed and produced nothing.
- **A learner follows a deep link into a passage whose time span exceeds twenty seconds.** The measured maximum is 20.22 s. Landing “inside the segment” is the guarantee; landing on the word

is not, and the interface must not imply the precision it does not have.

- **The taxonomy is re-extracted while the index is being served.** Half-written taxonomies must never be served; a stale but coherent taxonomy is preferable to an incoherent one.
- **A search hit's containing unit was deleted between indexing and display.** The locus must fail loudly rather than render a link into nothing.
- **An area has zero questions.** Presenting an empty practice set as though practice exists is a small lie that erodes trust in every other count on the page.
- **A screen-reader user follows a deep link into the recording.** The jump changes context; the change must be announced, and returning must restore focus, not dump the user at the top of the page.

Requirements (*mandatory*)

Functional Requirements

Ported knowledge model — port, do not clone

- **FR-001:** The workshop **MUST** adopt the reference module's learning hierarchy — curriculum, track, module, lesson — as its curriculum spine, taking the shape from the reference rather than inventing a second one.
- **FR-002:** The mapping between that hierarchy and the workshop's existing concepts — chapters, passages, cross-references — **MUST** be defined explicitly and **MUST NOT** create a parallel hierarchy that covers the same ground. A knowledge area is the workshop's module; a chapter is a *source* of material for areas, not a level of the hierarchy above them.
- **FR-003:** For every element taken from the reference, the design **MUST** record which of three it is — **ported as-is**, **adapted**, or **newly designed** — and why. An element recorded as “ported as-is” that in fact differs is a defect.
- **FR-004: The reference supplies shape, never substance.** No content, wording, fixture, question, model answer or asset **MAY** be copied from the reference module into the workshop. What is reused is structure: the hierarchy, the document skeleton, the question-bank schema, the export toolchain's design. What is authored fresh is everything a reader actually reads. Every workshop area, lesson and question **MUST** be derived from the

workshop's own chapters and MUST be evidenced against the workshop's own passages. *Why this is stated so bluntly*: the reference holds 785 ready-made questions and 34 finished area documents about a different subject entirely. Reusing them would be faster than authoring, would pass a superficial review because the documents look right, and would be worthless — the workshop would ship a curriculum about interview preparation while claiming to teach its own material. It is the single easiest way for this feature to go wrong.

- **FR-004a**: The reference module is a **separate private repository**. Its content MUST NOT migrate into the workshop under any justification, including as a placeholder, a fixture, a template with the words left in, or an example. This is a content boundary, not a style preference, and it is enforced by the same check that keeps workshop content out of public repositories.
- **FR-005**: The normative definition of the ported model MUST live in a single knowledge-model contract document, and this specification and every downstream artifact MUST reference it rather than restate it, so that the two cannot drift.

Taxonomy extraction

- **FR-006**: The system MUST derive its knowledge areas and terms from the indexed corpus, not from a hand-written list, and MUST record how each was derived.
- **FR-007**: Every area and every term MUST carry a persisted identifier assigned once at creation, neither positional nor derived from its own text, following the same identity discipline the passage identifier already follows.
- **FR-008**: Every published area and every published term MUST be evidenced by at least one passage. An area or term with no evidencing passage MUST NOT be published.
- **FR-008a**: The five areas that already exist as prose in the workshop's training documentation MUST be **promoted** into the machine-readable model — given identifiers, evidenced against the corpus and reconciled with what extraction derives — rather than being ignored, duplicated, or overwritten by a fresh extraction. Where extraction and the existing prose disagree about what an area is, the disagreement MUST be surfaced for a decision, not silently resolved in either direction.
- **FR-009**: Extraction MUST be idempotent: a second run over unchanged content MUST produce an identical taxonomy and MUST change no identifier.

- **FR-010:** Extraction MUST account for every passage — each is either attached to at least one area or explicitly classified as unattached with a stated reason. Silent omission is forbidden.
- **FR-011:** The system MUST distinguish a term identified with confidence from one identified with low confidence, MUST mark the latter, and MUST NOT present it as the former.
- **FR-012:** An area evidenced only by passages flagged uncertain MUST be marked as resting on uncertain evidence.
- **FR-013:** Term significance MUST be demonstrated by a recorded, re-derivable measure rather than asserted, and raw frequency alone MUST NOT be that measure.

Area materials

- **FR-014:** Every published area MUST carry a materials document following the reference module's section skeleton — measured as seven level-2 sections in a fixed order: why the area matters, its core concepts on one page, questions and model answers subdivided per topic, real-world challenge scenarios, red flags and strong signals, how the area maps back to the workshop, and sources. The two sections whose reference wording is specific to interviewing MUST be **adapted** to the workshop's subject rather than carried over verbatim, and that adaptation MUST be recorded under FR-003.
- **FR-014a:** Where an area's materials contain a diagram, the diagram MUST be authored as text source and rendered, never embedded as an unreproducible binary, following the reference module's practice.
- **FR-015:** Every substantive claim in an area's materials MUST carry at least one passage citation that resolves, or MUST be visibly marked as editorial framing that is not workshop content.
- **FR-016:** An area MUST NOT be published until its publication review has been recorded. Recording "*nothing to change*" is a valid review; skipping the review is not.
- **FR-017:** Materials MUST record for each unit of prose whether it was authored or assembled, and MUST NOT present assembled text as authored.

Deep linking — the identity and navigation model

- **FR-018:** Every relationship the system records between two addressable things MUST be traversable in both directions. A relationship that can only be followed one way MUST NOT be recorded.

- **FR-019:** From any transcript moment, the system **MUST** return the areas, the lesson section and the questions attached to it, each reachable in a single step.
- **FR-020:** From any area, lesson section or question, the system **MUST** return every passage that evidences it, with the position at which the evidence occurs.
- **FR-021:** For evidence that is media-backed, the position **MUST** be a time span, and following it **MUST** land within that span. Two precisions exist and the system **MUST NOT** conflate them: **word level**, available where a word-timing sidecar covers the material, and **segment level**, available everywhere else. Every media-backed link **MUST** record which precision produced it, **MUST NOT** present segment precision as word precision, and **MUST NOT** treat a word whose timing confidence is below the recorded threshold as though it were confident.
- **FR-022:** A mention that spans a segment boundary **MUST** be attached to every segment it touches.
- **FR-023:** Every link **MUST** resolve through one shared resolution path with four distinct outcomes — found, redacted, not present, and could not determine — with no fuzzy fallback, no nearest-match, and no second path.
- **FR-024:** An unresolvable link **MUST** fail loudly and **MUST NEVER** be silently re-pointed at a different target.
- **FR-025:** All links **MUST** continue to resolve to their original targets after a passage's text is corrected and the content re-indexed.
- **FR-026:** Traversal of the relationship graph **MUST** terminate on cyclic material and **MUST NOT** trap a reader.
- **FR-027:** Redaction **MUST** propagate to every artifact kind this feature introduces — taxonomy, materials, questions, index entries, exports and stored answers — not only to displayed transcript.

Search

- **FR-028:** Every content kind the feature introduces — areas, lesson sections, terms and questions — **MUST** be indexed and searchable alongside the existing kinds.
- **FR-029:** Every search result **MUST** carry a locus: the unit that contains it and the position within that unit. The position **MUST** include the **match offsets within the matched text**. Today the result already advertises an offset list and that list is always empty; a field that promises a position and never carries one **MUST NOT** survive this feature. Either it is populated, or it is removed and its absence stated.
- **FR-030:** A locus **MUST** resolve. A result whose locus cannot be resolved **MUST NOT** be returned as though it could.

- **FR-030a:** The reference module’s search **MUST NOT** be ported. It is a substring scan with no ranking and no locus, and the workshop’s existing search already exceeds it on both counts. This is recorded as an explicit “not ported” decision under FR-003 so that a later reader does not mistake the omission for an oversight.
- **FR-031:** Suggestions **MUST** include area and term names.
- **FR-032:** Results **MUST** be filterable by content kind and by area.
- **FR-033:** The existing three-state search behaviour — results, no relevant match, could not determine — **MUST** be preserved across every new kind. A degraded backend **MUST NOT** be rendered as an empty result set.

Full connectivity across all four content kinds

- **FR-033a:** Four content kinds **MUST** be interlinked, and the requirement is not satisfied by linking any subset: **transcripts, the recording, codebases, and the knowledge base** — areas with their lessons and their questions. Every pair that can meaningfully relate **MUST** be traversable in both directions:

From	To	What the traversal must return
a moment in the recording	area / lesson / question	what is being taught at that moment and what tests it
an area, lesson or question	a moment in the recording	the position where the material was actually taught
a transcript passage	area / lesson / question	the knowledge the passage evidences
a code passage	transcript passage	the moment that code was discussed
an area	code passages	the code that the area’s knowledge is realised in
any of the four	any other, transitively	a path, not a dead end

- **FR-033b:** The join across all four kinds **MUST** be the passage identifier. A timestamp **MUST NOT** be used as a join key. *Why:* the identifier is measured to survive both a content correction and a

structural insertion; a timestamp survives neither a re-cut nor a re-transcription, so a graph keyed on time would silently rot at the first correction while continuing to render links.

- **FR-033c:** Search MUST cover all four kinds by meaning and by context, and MUST report where each hit occurs, in the terms of the kind it came from — a time position for the recording, a text position for a transcript or a document, a file and symbol position for code, and a containing area and lesson for the knowledge base.

A repeatable chapter pipeline, not a one-off build

- **FR-033d:** Everything this feature produces MUST be produced by a **repeatable pipeline**, not hand-assembled for the chapter that exists today. Processing a new chapter MUST yield, without hand-authoring of the structure: its transcript, its passages with minted identifiers, its extracted areas and themes, its authored materials, its long and short question sets, its cross-references, its search index entries and all of its deep links.
- **FR-033e:** That pipeline MUST **extend the chapter-addition path the platform already contracts** — the documented add-a-chapter procedure and its accompanying prompt — rather than introducing a second, parallel way to add a chapter. Two procedures for the same act guarantee that one of them rots.
- **FR-033f:** Adding a chapter MUST update the taxonomy rather than replace it: areas already established keep their identifiers and gain new evidence, areas newly evidenced are added, and an area that a new chapter contradicts is surfaced for a decision rather than silently rewritten.
- **FR-033g:** The pipeline MUST report precisely what is missing when a chapter cannot be fully processed, and MUST NOT publish a partially processed chapter as though it were complete.

Assessment with provenance

- **FR-034:** The system MUST support the reference module's question kinds — multiple choice, short answer and flashcard — with the reference's difficulty and category vocabulary, taking the schema from it and the questions from the workshop's own material.
- **FR-034a:** Every area MUST carry both a **short** question set and a **long** question set, and the distinction MUST be a real difference in depth rather than in count: a short set checks recall of the area's core concepts, a long set requires reasoning across more than one passage of evidence.

- **FR-034b:** Question coverage **MUST** be **exhaustive against the area's evidence, not against a target count**. Every lesson section of an area **MUST** be assessed by at least one question, and the proportion of the area's evidencing passages that are cited by at least one question **MUST** be measured and published. A large number of questions concentrated on a small part of an area does not satisfy this requirement, and the measurement is defined so that it cannot.
- **FR-035:** Every question **MUST** carry at least one passage citation. This is the property the reference module does not have, and it is mandatory here.
- **FR-036:** A question whose citation does not resolve **MUST NOT** be served.
- **FR-037:** From any question, a learner **MUST** be able to reach the position its citation names and return, with their progress preserved.
- **FR-038:** A model answer that was assembled rather than authored **MUST** be marked as assembled.
- **FR-039:** The system **MUST** record a learner's progress per area and per question set, and that record **MUST** survive a re-ingest of the content it refers to — a learner's history is not a derived artifact and **MUST NOT** be discarded when the curriculum is rebuilt.

Learning surface and visual identity

- **FR-040:** The interface **MUST** offer the reference module's learning views — an area view, a practice view, a progress view and a study-plan view — as first-class surfaces.
- **FR-041:** The workshop's existing chapter, transcript, recording and cross-reference capabilities **MUST** remain reachable, subordinated to the area they serve rather than removed.
- **FR-042:** All new styling **MUST** resolve through the shared design-token contract, and **MUST NOT** write brand values literally.
- **FR-043:** The workshop **MUST** have a visual identity distinguishable from the reference module, achieved through the token layer rather than by forking the component library.
- **FR-044:** Every new view **MUST** meet WCAG 2.1 Level AA, and every deep link **MUST** be operable by keyboard alone, with context changes announced.
- **FR-044a:** Interaction, motion and transition behaviour **MUST** be taken from the platform's own ranked, sourced user-experience research rather than chosen by taste, and each interaction decision **MUST** name the finding it rests on. Where a proposed effect has no supporting finding, it **MUST NOT** ship on the grounds that it looks better.

- **FR-044b**: Every motion effect **MUST** respect a reduced-motion preference, preserving the final visual state without the transition. An effect that becomes load-bearing for comprehension — one a user who disables motion would lose information by not seeing — **MUST NOT** be used.

Export

- **FR-045**: Every published area **MUST** be exportable to Markdown, HTML, Word and PDF.
- **FR-046**: The non-Markdown formats **MUST** be generated from the Markdown source; no format may be authored independently of the others.
- **FR-047**: Citations **MUST** be preserved in every exported format.
- **FR-048**: An area that has not passed publication review **MUST NOT** be exported, and the omission **MUST** be reported.
- **FR-049**: When the export toolchain is unavailable, the system **MUST** report that export could not be performed, and **MUST NOT** report success over an empty or partial output.

Answering over the new material

- **FR-050**: The known unclosed defects in question answering — the measured fabrication rate on unanswerable questions, and the undefended case where a topically related fragment that does not answer the question passes every existing check — **MUST** be stated in the shipped limits documentation, and **MUST NOT** be described as solved anywhere.
- **FR-051**: If answering is extended to generate answers about areas, lessons or questions, a verification layer that checks the answer against the **question** — not only against the passage — **MUST** exist and **MUST** be measured before that extension is enabled.
- **FR-052**: Where a verification layer cannot run because its prerequisite is absent on the host, it **MUST** report that it could not determine, and the answer **MUST NOT** be served as verified. An entailment-based verifier already exists in the platform behind a configured model directory, with lexical overlap as its fallback; a verifier that was *requested* and could not load **MUST NOT** silently degrade to the weaker one.

Evidence and governance

- **FR-053**: Every check this feature adds **MUST** have a paired demonstration that it reports failure when the condition it guards is broken.

- **FR-054:** Every check MUST distinguish “could not determine” from both “passed” and “failed”.
- **FR-055:** Checks MUST write their evidence to the repository’s versioned evidence location.
- **FR-056:** No server-side continuous-integration automation may be introduced, anywhere in the fleet.
- **FR-057:** No workshop content — no transcript text, no participant name, no chapter material, no recording filename — may appear in any public repository, including this specification directory. Content is referenced by identifier only.
- **FR-058:** Components this feature produces MUST follow the decoupling and reuse rules already established for the workshop platform: usable without the curriculum, testable without it, and where genuinely reusable, resident in their own public repository with synthetic fixtures.
- **FR-059:** Every endpoint this feature adds MUST be declared in the platform’s interface contract **and** in the route manifest the server-unity verifier reads, and MUST be probed against the running binary. The existing verifier already fails when a contracted endpoint has no manifest row, so an endpoint that is built but undeclared is a failure by construction — and one that is declared but not built must answer as a not-built route rather than being swallowed by the single-page application.

On-screen text — added 2026-09-02 by the decision recorded in D5

Every requirement in this block is **specified and unbuilt**. They are carried by tasks T123–T142, sequenced after this specification’s 28 already-unbuilt tasks.

- **FR-060:** On-screen text recognised from a chapter recording MUST enter the corpus as a **passage** of the new kind `screen_text`, minted through the **same** minter as every other passage and resolved through the **same** four-outcome resolver. It MUST NOT be stored in a second registry, mint through a second minter, or carry a second identifier format.
- **FR-061:** Each `screen_text` passage MUST carry a **visibility interval** — the span across which its text was observed on screen — and the **interval bound**, which is the sampling period the interval is accurate to. The bound MUST be recorded on the passage and carried onto every mention derived from it. It MUST NOT be omitted, and it MUST NOT be presented as zero.
- **FR-062:** A mention derived from a `screen_text` passage MUST join the **existing two-valued** precision model: it declares segment precision and carries its interval bound. **No third precision value**

may be introduced. A `screen_text` mention MUST NOT declare word precision, because no per-word timing record produced its time.

- **FR-063:** Every mention MUST declare its **modality** — spoken or `on_screen`. Every evidence count on which a publication, coverage or attachment figure depends MUST be taken over **corroboration groups** and not over raw mentions, so that one term both spoken and displayed at one moment counts **once**. Both figures — grouped and raw — MUST be published together. Neither mention may be deleted: both MUST remain individually retrievable and individually navigable.
- **FR-064:** OCR accuracy MUST be **measured per chapter**, against a hand-truthed ground-truth sample drawn by a recorded, seeded, reproducible rule with the population size published, on **both** axes: **textual** accuracy and **temporal** accuracy — whether a mention's declared visibility interval actually contains the moment its text was on screen. A textual figure alone does not satisfy this requirement, because deep linking depends only on the temporal one and a text-only measurement cannot see a temporal failure at all.
- **FR-064a:** The accuracy **budget** MUST be **derived from the recorded speech-recognition calibration for the same corpus and read from it at run time**, never written as a literal. It MUST NOT be a guessed threshold. Where the calibration record or the ground-truth sample cannot be read, the check MUST report **could not determine** and MUST NOT pass.
- **FR-064b:** No OCR-derived mention may be **published** for a chapter until that chapter's accuracy measurement has been observed passing **for that chapter**. Font, resolution and compression differ between recordings, so an accuracy figure measured on one chapter is not evidence about another — the same per-chapter discipline word-timing availability already carries.
- **FR-065:** OCR output MUST pass the content-boundary check **before** it reaches any public artifact. A screen recording displays material the transcript never contained — window titles, file paths, identifiers, a third party's name in a title bar — so this kind is a **new disclosure surface**, and a boundary check scoped to transcript text does not cover it.
- **FR-066:** The `screen_text` kind MUST NOT be advertised as an indexed kind until it is **retrievable**, proven by a planted known target rather than by a row count. The corpus already advertises one kind with zero entries, and this requirement exists so that a second one is not created.

Key Entities

- **Knowledge Area:** a coherent body of knowledge the workshop covers. The workshop's realisation of the reference module's *module*. Carries a persisted identifier, a title, a summary, an ordering, its evidencing passages, its lesson sections, its questions and its publication-review record. An area is *not* a chapter and does not live inside one; it draws evidence from any number of chapters.
- **Term:** a named concept within one or more areas. Carries a persisted identifier, a canonical form, its known surface variants, a confidence marking and its evidencing passages. A term with no evidence does not exist.
- **Lesson Section:** an addressable part of an area's materials. Carries a persisted identifier and its own citations, so a link can name a part of an area rather than only the whole.
- **Question:** an assessment item belonging to an area. Carries a persisted identifier, its kind, its prompt, its answer or choices, an authored-or-assembled marking, and — mandatorily — its citations. **This last field is the difference from the reference module, where it does not exist.**
- **Mention:** the join that carries the whole feature. Relates a term or an area to a passage, and records the position within that passage — a text span always, and for media-backed passages the containing segment's time span. Mentions are what make both directions of navigation possible.
- **Passage:** unchanged from the existing platform. The smallest addressable unit of source content, carrying a minted identifier that is neither positional nor content-derived. This feature adds nothing to it and depends on it entirely.
- **Publication Review:** the record that an area has been reviewed and may be published and exported. Has a subject, a decision, a reviewer and a time. Its absence blocks publication.
- **Locus:** what a search result reports about where it is — the containing unit and the position within it. Distinct from the result itself, because the same passage can be reached through more than one containing unit.
- **Progress:** a learner's state within an area and within a question set.
- **On-Screen Text Passage** (`screen_text`): a passage whose text was **shown** rather than spoken, read from a chapter recording by optical character recognition. A new **kind** of the existing Passage, not a new entity — it inherits the minter, the resolver, the redaction flag and the mention model unchanged. Carries a visibility interval

and the interval bound that interval is accurate to. **Added 2026-09-02; specified and unbuilt.**

- **Corroboration Group:** the unit evidence is **counted** over. Groups the mentions of one subject whose times overlap within a measured window, so a term both spoken and displayed at one moment is one piece of evidence rather than two. It changes what is counted, never what exists — every grouped mention stays individually retrievable. **Added 2026-09-02; specified and unbuilt.**

Success Criteria (*mandatory*)

Measurable Outcomes

Every criterion below names the method that measures it. A criterion whose measurement method is “inspection by a reviewer” says so and states the sample size and how the sample is drawn. This discipline exists because feature 001 shipped a criterion that no task measured, and the gap was found late.

Taxonomy

- **SC-001:** 100% of published areas and 100% of published terms have at least one evidencing passage. **Measured by:** joining the taxonomy to the mention set and counting entries with zero mentions; the count must be 0. Paired mutation: publish an area with its mentions removed — the check must go red.
- **SC-002:** On a random sample of at least 30 extracted terms, drawn with a recorded seed, at least 90% are judged genuine domain terms rather than extraction noise. **Measured by:** a reviewer’s recorded judgement sheet, stored with the seed so the sample is reproducible and the verdict is challengeable.
- **SC-003:** 100% of the 1,101 passages are accounted for — attached to at least one area or classified as unattached with a stated reason. **Measured by:** summing attached and classified counts and asserting the total equals the registry count. The attachment *proportion* is published as a measured figure; no threshold is claimed for it, because the corpus’s topical density has not been measured and inventing a threshold would be a guess.
- **SC-004:** A second extraction run over unchanged content changes nothing: 0 identifiers reassigned, 0 areas or terms added or removed, and a byte-identical taxonomy file. **Measured by:** running extraction twice and diffing; paired mutation: make one identifier depend on content — the check must go red.

Materials

- **SC-005:** 100% of published areas contain every section of the reference skeleton. **Measured by:** a heading-presence check across all area source documents; any missing section fails.
- **SC-006:** 100% of substantive claims in published areas either carry a resolving citation or are marked as editorial. **Measured by:** parsing each area document into claims, resolving every citation, and asserting no claim is both unmarked and uncited.
- **SC-007:** 0 areas are published without a recorded publication review. **Measured by:** a publication precondition check that enumerates published areas and asserts each has a review that is newer than the materials it reviews.

Deep linking

- **SC-008:** For 100% of recorded relationships, reverse traversal returns the origin. **Measured by:** enumerating every relationship, traversing it in reverse through the platform's own interface, and counting failures; the count must be 0.
- **SC-009:** For 100% of media-backed citations, following the link lands inside the cited passage's own time span. **Measured by:** asserting the landing time falls within the span, over every such citation. The precision this delivers is the segment, whose measured distribution is median 6.74 s, p95 10.78 s and maximum 20.22 s; that distribution is published with the result rather than being smoothed into a single number.
- **SC-010:** After a content correction and a structural insertion, 100% of area, term, lesson, question and mention links still resolve to the same targets. **Measured by:** replaying the identifier-survival experiment already measured for passages, extended to the new identifier kinds, and comparing the full link set before and after by hash.
- **SC-011:** 100% of unresolvable links fail loudly; 0 are silently re-pointed. **Measured by:** a mutation that deletes a link target and asserts the resolver returns a loud outcome; the paired proof makes the resolver re-point by nearest match and requires the check to go red.
- **SC-012:** 100% of redactions propagate to every artifact kind this feature introduces. **Measured by:** redacting a passage and asserting its text is absent from the taxonomy, the materials, the questions, the index, the exports and any stored answer; paired mutation: skip one propagation target — the check must go red.

Search

- **SC-013:** 100% of content kinds are represented in the index.
Measured by: for each kind, a query with a planted, known target must return it; a kind with content but zero index entries fails. This is deliberately not a count of rows, because a row count cannot tell a populated index from a populated table nobody queries.
- **SC-014:** 100% of returned search hits carry a locus that resolves.
Measured by: running the retrieval benchmark and asserting the property on every hit; any hit without a resolving locus fails the run.
- **SC-015:** On a benchmark of at least 20 meaning-based queries over the new content kinds, with expected targets fixed in advance, at least 90% return the expected item in the top five. **Measured by:** a stored benchmark file of query-to-expected-identifier pairs, executed by a gate that prints the per-query outcome, not only the aggregate.
- **SC-016:** Suggestion latency remains at or below 200 ms at the 95th percentile after the new kinds are indexed. **Measured by:** re-running the existing latency harness against the grown corpus and publishing the measured p95 alongside the pre-existing figure, so a regression is visible rather than merely absent.
- **SC-017:** Search result latency remains at or below 2 s at the 95th percentile after the new kinds are indexed. **Measured by:** the same harness and the same before-and-after publication.

Full connectivity and the repeatable pipeline

- **SC-015a:** All six traversals in the connectivity matrix are exercised and return a non-empty result wherever the underlying material exists. **Measured by:** a connectivity harness that, for each row of the matrix, picks every eligible origin and asserts the traversal returns at least one target — and, where no target exists, asserts the system says so rather than returning an empty list that reads as “nothing here”. Any row with zero exercised origins fails, because an unexercised traversal is unmeasured, not passing.
- **SC-015b:** 0 links in the knowledge graph are keyed on a timestamp. **Measured by:** inspecting every stored relationship for its join key; any relationship keyed on time rather than on a passage identifier fails. Paired mutation: introduce one time-keyed link — the check must go red.
- **SC-015c:** A chapter that has never been processed goes from raw material to a complete, connected knowledge layer using only the documented procedure, with **zero hand-authoring of structure and zero code changes**. **Measured by:** running the procedure end

to end against a small synthetic chapter and asserting the presence of every output — passages with identifiers, areas, materials, short and long question sets, cross-references, index entries and deep links — then asserting the diff contains no manually created structural file. This is the criterion that proves the pipeline is repeatable rather than that one chapter happens to be finished.

- **SC-015d:** Adding a chapter preserves the existing taxonomy: 0 identifiers of previously established areas or terms change, and every contradiction between new evidence and an existing area is reported rather than resolved silently. **Measured by:** capturing the taxonomy before and after, diffing identifiers, and asserting the contradiction report exists and is non-empty whenever a contradiction was seeded. Paired mutation: make the pipeline re-derive area identifiers on each run — the check must go red.
- **SC-015e:** An incompletely processed chapter is never published as complete. **Measured by:** withholding one required input, running the procedure, and asserting it names precisely what is missing and publishes nothing. Paired mutation: make the missing input a warning — the check must go red.

Assessment

- **SC-017a:** 100% of areas carry both a short and a long question set, and the two are distinguishable by depth rather than by count. **Measured by:** asserting both sets exist for every area, then asserting that every question in a long set cites **more than one distinct passage** while a short-set question may cite one. The depth test is structural on purpose — a reviewer's impression of difficulty is not reproducible, and a citation count is.
- **SC-017b:** 100% of lesson sections in every published area are assessed by at least one question. **Measured by:** joining lesson sections to the questions that cite passages belonging to them and counting sections with zero; the count must be 0.
- **SC-017c:** The proportion of each area's evidencing passages that are cited by at least one question is **measured and published per area**, and the distribution across areas is published with it. **Measured by:** computing the ratio per area and emitting the full per-area table, not an average. No threshold is claimed, for a stated reason: the density of assessable substance per passage has not been measured, and inventing a percentage would be a guess dressed as a target. What the criterion forbids is the figure going unmeasured or being reported only as a mean, which is how a well-covered area conceals an empty one.

- **SC-018:** 100% of served questions carry at least one citation that resolves. **Measured by:** enumerating every served question, resolving every citation, and counting failures; the count must be 0. **The reference module's measured value for this property is 0%,** so this criterion is where the workshop exceeds it, and the comparison is recorded rather than claimed.
- **SC-019:** 100% of questions whose citation is a transcript segment support a round trip — from question to the cited moment and back — with progress preserved. **Measured by:** an end-to-end traversal over every such question; if the count of such questions makes exhaustive traversal impractical, a sample of at least 30 is drawn with a recorded seed and the total population is published alongside it.
- **SC-020:** 100% of assembled model answers are marked as assembled. **Measured by:** a schema assertion over the whole question set; an unmarked assembled answer fails.

Surface

- **SC-021:** Every new view reports zero Level A and zero Level AA accessibility violations, and 100% of deep links are operable by keyboard alone. **Measured by:** an automated audit per view plus a keyboard-only traversal that visits every deep-link affordance and asserts it can be reached, activated and returned from.
- **SC-022:** 100% of new styling resolves through the shared design-token contract; 0 literal brand values appear in new stylesheets. **Measured by:** the design toolkit's own token conformance check, plus a literal-value scan over the new stylesheets; both must pass, because the first proves the tokens are valid and the second proves they are actually used.
- **SC-023:** 100% of the existing chapter, transcript, recording and cross-reference capabilities remain reachable after the surface is reshaped. **Measured by:** enumerating those capabilities before the change and asserting each is still reachable after it — a before-and-after list, not a recollection.

Export

- **SC-024:** 100% of published areas carry all four formats. **Measured by:** a file-existence matrix over every published area. The reference module's measured coverage is 25 of 34; this criterion requires the workshop to be complete, and the comparison is stated so the bar is known to be higher than the model it was taken from.

- **SC-025:** Re-exporting an unchanged area produces documents whose extracted text is identical to the previous run. **Measured by:** extracting text from each output and diffing; extraction rather than byte comparison, because embedded timestamps make byte equality unachievable for some formats and a criterion nobody can meet is worse than none.

Honesty and governance

- **SC-026:** The shipped limits documentation names every open defect the evidence directory records, including the measured fabrication rate on unanswerable questions and the undefended topically-related-but-non-answering case. **Measured by:** a cross-check that reads the recorded defects and asserts each is named in the limits document; an unnamed defect fails. Paired mutation: remove one defect from the limits document — the check must go red.
- **SC-027:** 100% of checks this feature adds have a paired demonstration that they fail when their guarded condition is broken. **Measured by:** the check registry, which fails on an unregistered or unpaired check.
- **SC-028:** 100% of checks this feature adds distinguish “could not determine” from “passed” and “failed”. **Measured by:** driving each check into a could-not-determine condition — an absent dependency or an unreachable service — and asserting the third state rather than either of the other two.
- **SC-029:** 0 occurrences of workshop content appear in any public repository, including this specification directory. **Measured by:** the content-boundary check, run across the public surface.
- **SC-029a:** 0 occurrences of reference-module content appear anywhere in the workshop. **Measured by:** the same boundary check run in the opposite direction — every workshop area, lesson and question is checked against the reference’s own corpus, and any substantive overlap fails. Paired mutation: plant one reference question in a workshop bank — the check must go red. This runs in both directions deliberately: one direction protects a private recording from a public repository, the other protects the workshop from shipping someone else’s curriculum under its own name.
- **SC-029b:** 100% of interaction and motion decisions in new views name the research finding they rest on. **Measured by:** a traceability check over the interaction inventory; an entry with no named finding fails.
- **SC-030:** 0 active server-side continuous-integration workflows exist anywhere in the fleet after this feature lands. **Measured by:** the existing fleet-wide gate.

On-screen text — added 2026-09-02, and the reason this feature cannot close on design alone

Every criterion in this block is **specified and unmeasured**. SC-031 through SC-034 are the ones that make “*spec 002 cannot close until OCR accuracy is measured*” a checkable statement rather than a sentiment: each names a figure that does not exist yet, and none of them can be satisfied by building the pipeline — only by running it and publishing what it produced.

- **SC-031:** OCR **textual** accuracy is measured and published per chapter, as word error rate and character error rate against a hand-truthed ground-truth sample. **Measured by:** a ground-truth sample drawn by a recorded seed with the population size published, scored by the same edit-distance method the speech-recognition calibration already uses, so the two figures are comparable rather than merely adjacent. A published figure with no recorded seed fails.
- **SC-032:** OCR **temporal** accuracy is measured and published per chapter: the proportion of on-screen mentions whose declared visibility interval contains the frame at which the text was actually on screen. **Measured by:** resolving each sampled mention back to the recording and asserting containment. **This criterion is separate from SC-031 deliberately** — deep linking depends on this figure and on no other, and a perfect textual score is compatible with every interval being wrong.
- **SC-033:** Both OCR figures are published **beside** the speech-recognition figures for the same corpus, in the same document, in the same run. **Measured by:** a check that reads the recorded speech calibration and asserts both appear together; an OCR figure published alone fails, because a number with nothing to compare it to is not an accuracy claim a reader can act on.
- **SC-034:** 0 OCR-derived mentions are published for any chapter whose accuracy measurement has not been observed passing **for that chapter**. **Measured by:** joining the published mention set to the per-chapter accuracy records; a mention from an unmeasured or could-not-determine chapter fails. Paired mutation: publish from a chapter whose accuracy run returned could-not-determine — the check must go red.
- **SC-035:** A term both spoken and displayed at one moment contributes **1** to its area’s evidence count, not 2, while **both** of its mentions remain individually retrievable and individually navigable. **Measured by:** seeding exactly that case and asserting the count, the retrievability of both members, and that the grouped and raw figures are both published. Paired mutations: count raw

mentions instead of groups; and widen the grouping window to the whole chapter so two genuinely distinct occurrences collapse. Both must go red.

- **SC-036:** 0 occurrences of on-screen-derived text reach a public artifact without passing the content-boundary check. **Measured by:** the existing boundary check, extended over the OCR output and run before publication. Paired mutation: scope the check to transcript text only — it must go red, because that is precisely the scoping that would let a window title through.

Assumptions

Decisions taken where the description did not specify, each chosen for the safest and most stable option and each revisitable.

- **The reference is a model, not a source.** Its structure, hierarchy, question kinds, document skeleton and export toolchain are the template; none of its content, wording or fixtures crosses into the workshop. Its 34 finished area documents and 785 finished questions are about a different subject and are, for this feature's purposes, not material but a shape.
- **Everything a reader reads is authored from the workshop's own chapters.** Areas, lessons and questions are derived from what was actually taught and evidenced against the workshop's own passages. There is no scenario in which borrowed material is acceptable, including as a temporary placeholder.
- **This is a pipeline, and it is judged as one.** One chapter exists today. The feature is complete when *the next* chapter can go through it and come out with the same completeness, not when this chapter's materials look finished.
- **A knowledge area is the workshop's module.** Mapping areas onto the reference's *module* level keeps one hierarchy rather than two, which FR-002 requires. Chapters supply evidence to areas; they are not a level above them.
- **Areas span chapters from the first day.** The corpus currently has one chapter, so a one-chapter-per-area shortcut would work today and break at the second chapter. Modelling the relationship as many-to-many now costs nothing and avoids a migration later.
- **The taxonomy is bounded by evidence, not by the reference's size.** The reference has 34 areas. The workshop will have as many as its corpus evidences and no more. Producing areas to match a count would mean inventing coverage the recording does not contain — the exact failure the governing rules forbid.

- **Identity discipline is inherited wholesale.** Areas, terms, lesson sections, questions and mentions all take minted, persisted identifiers on the same rules as the passage identifier, because the property that makes links survive corrections is already measured for passages and there is no reason to invent a weaker scheme for the things pointing at them.
- **Mention precision is the containing segment.** Word-level timings do not exist in the registry, so segment-level is what can be delivered honestly. The measured distribution is published rather than a single reassuring number.
- **Assessment mirrors the reference's grading model.** Multiple choice is scored automatically; short answer reveals a model answer for self-assessment. Automatic grading of free text would require a capability the host does not have and would import the fabrication risk into the one place a learner is most likely to trust the output.
- **The corpus for area extraction is the workshop's own material.** Transcript segments, chapter documents and diagrams create areas. Code passages in the wider monorepo may be *linked* as evidence, following the corpus decision already recorded for the platform, but they do not create areas — code in an unrelated project is not a thing the workshop taught.
- **The platform stays local and internal.** The publication scope already decided for the platform is unchanged; nothing here implies public hosting, and the privacy handling and redaction path already contracted continue to apply.
- **Search is extended, not replaced.** The existing two-path search and its three-state behaviour are proven; new kinds join that index rather than arriving with a second search system.
- **The existing front end is reshaped, not rebuilt.** It has more code than the reference and it works; the defect is its shape. Subordinating the media views under areas preserves working behaviour while fixing the shape, and avoids discarding capability that took a feature to build.
- **The visual identity comes from the token layer.** A distinct look is achieved by giving the workshop its own token values against the shared contract, not by forking the component library — forking would double the maintenance and guarantee drift.
- **Scale is tens of areas and one to tens of chapters,** matching the platform's existing scale assumption. Nothing is designed for thousands.
- **Governance is binding.** No server-side CI, paired mutation proofs for every gate, three-valued checks, evidence written to the

versioned location, carriers in lockstep, and the continuation document updated alongside non-trivial changes.

Dependencies

- **The knowledge-model contract** in the workshop's documentation. It is the normative definition of the ported hierarchy, and this specification deliberately references it rather than restating it (FR-005), so the two cannot drift. **It landed while this specification was being written**, and its opening rule — that the contract transfers *form* and transfers no *content* — is the same boundary FR-004 states independently. The two were written in parallel and agree; that agreement is a check that passed, not a coordination.
- **The area and term taxonomy**, extracted into the workshop's curriculum directory by a separate effort. **It also landed while this specification was being written**, and the requirements here were written against what a taxonomy must *satisfy* rather than against any particular one — so what landed is evidence, not a premise. Measured on arrival, for the one chapter that exists: **7 tracks, 37 modules, 137 terms**; per-passage linkage for all **1,101** passages; a coverage record showing **884 of 1,055 transcript passages covered (83.79%)** and **14,290 of 15,610 words (91.54%)**, with every uncovered passage carrying a **stated reason**; and a reverse index keyed by module, by term and by track. Its own identity rule states that the passage identifier is the citable identity and that timestamps are for navigation only because a re-cut moves them — which is FR-033b, arrived at independently. *What this does not settle*: the taxonomy is an input to be reconciled under FR-008a and contract §2.3, not a finished answer. The five prose areas still hold the promotion path, and any contradiction between them and these 37 modules is reported for a decision rather than merged.
- **The platform's user-experience research** (FR-044a). **This has not landed**. The research directory now exists but holds a document on a different subject. Until it does, the interaction and motion requirements state what a decision must be able to show, and **T094 is marked blocked** rather than being written against taste.
- **The existing platform**: the passage registry and its minted identifiers, the resolution path, the search index and its three-state behaviour, the cross-reference graph, and the media range-serving that makes a timestamp jump possible.
- **The reference module**, as the structural template and as the source of the export toolchain's design — and **only** as that. It is a

separate private repository; it is read for shape and never for substance (FR-004, FR-004a).

- **The documented chapter-addition procedure and its prompt**, which exist in the workshop's documentation today while the script that automates them does not. This feature extends both rather than replacing them (FR-033e).
- **The platform's user-experience research**, being produced as this specification is drafted, as the source of the interaction and motion requirements (FR-044a). At the time of writing the research directory did not yet exist, so this specification states what interaction decisions must be able to show rather than naming any particular finding.
- **The shared design system and the design toolkit** that owns the token contract and its conformance check.
- **Document conversion tooling** for the four-format export.
- **An entailment-capable or generative model**, for the answer-to-question verification layer only, and only if the third clarification is resolved towards closing that defect. Nothing else in this specification depends on a model that this host does not have.

Out of Scope

- Public internet publication of the curriculum. The platform's local-and-internal decision stands.
- Re-transcribing, re-recording or editing source recordings.
- Authoring material for chapters that do not exist.
- Translating the curriculum.
- Replacing or re-architecting the reference module. It is read, never written.
- ~~Optical character recognition of on-screen content — pending the second clarification~~, which is precisely the question of whether this belongs in scope. **WITHDRAWN 2026-09-02: OCR of on-screen content is now IN SCOPE**, by the decision recorded in the Clarifications block and in **D5**. It is specified as tasks **T123–T142** and is **unbuilt**, sequenced after this specification's 28 already-unbuilt tasks. Do not restate this line as an exclusion — the pending question it referred to has been answered.
- Automatic grading of free-text answers.
- A second search system, a streaming service, or any new content delivery infrastructure.

Resolved Decisions

Questions that materially affect scope and were settled here rather than left open, each recording what was chosen, why it is the safe choice, and what would reopen it.

D1 — The knowledge area is the module level of the ported hierarchy

The question: the reference has curriculum, track, module and lesson. The workshop has chapters and passages. Where does a knowledge area sit?

Decision: a knowledge **area is the workshop's module**. Lesson sections sit beneath it. Chapters are a *source* of evidence that areas draw on, related many-to-many, and are not a level of the hierarchy.

Why this is the safe choice: the alternative — areas nested inside chapters — is simpler today, with one chapter, and wrong at the second one, because knowledge recurs across sessions. Choosing the correct shape while the data is small costs a design decision; choosing it later costs a migration of every link already created, which is exactly the class of change this feature's identity discipline exists to survive but should not be made to prove unnecessarily.

What would reopen it: a decision that areas are per-chapter by definition, which would make the curriculum a set of session summaries rather than a body of knowledge.

D2 — The existing media surface is subordinated, not removed

The question: the operator judged the current workshop poorly made. Do the existing chapter, transcript, recording and cross-reference views go away?

Decision: they **stay and are subordinated**. They become the way a learner reaches source material from within an area, rather than the top-level objects the product is organised around.

Why this is the safe choice: those views are the working half of what exists — they are what makes a timestamp jump possible at all, and the deep linking this feature is built on runs through them. Removing them would discard proven capability to fix a problem that is one of

arrangement. Keeping them without subordinating them would leave the product a media viewer with a curriculum bolted to the side, which is the criticism itself. Subordination fixes the shape and keeps the capability, and SC-023 measures that nothing was lost in the process.

What would reopen it: an explicit decision that the recording is not part of the learning experience.

D3 — Assessment is scored where scoring is objective, and self-assessed where it is not

The question: must the workshop grade free-text answers?

Decision: multiple-choice questions are **scored automatically**; short-answer questions reveal a model answer for **self-assessment**, as the reference module does. No automatic grading of free text.

Why this is the safe choice: automatic grading of free text needs a judging model. This host has none, so the capability would be unverifiable here — and worse, a grader is the single place where a wrong confident output does the most damage, because a learner told they were wrong when they were right will unlearn something correct. The reference module has lived without it. Deferring costs a feature; adding it now risks the product's credibility on a capability that cannot be measured on the machine it must run on.

What would reopen it: a judging model available and measured on this host, with a fabrication rate established the same way the answering path's was.

D4 — Citations are mandatory on questions, and a question without one is withheld

The question: what happens when a question cannot be traced to a passage?

Decision: it is **not served**. Provenance is a precondition of a question existing, not a decoration on it.

Why this is the safe choice: the entire justification for this feature over its reference is that the workshop's questions can be traced. A question served without provenance is indistinguishable to a learner from one

with it, so a single unprovenanced question silently converts the guarantee into a claim. Withholding is loud, cheap and reversible; serving is quiet and irreversible once trusted.

What would reopen it: nothing short of abandoning the provenance guarantee itself.

D5 — On-screen text is in scope, this spec is extended in place, and it is specified before it is built

Taken by the operator 2026-09-02. This is the decision that closes the *what “in videos” covers* clarification. It has three parts and each was decided separately, because each could have gone the other way.

The question: the description asks for terms mentioned “*in transcript and video*”. The corpus holds only what was **spoken**. Does “*video*” mean the recording as a source of spoken words, or does it mean the text visible on screen — the actual substance of a screen recording?

Decision, part 1 — scope: **spoken PLUS on-screen**. On-screen text is ingested by optical character recognition and becomes linkable evidence like any other passage. The alternative, spoken-only, was a linking feature over data that already exists and would have delivered a curriculum that could not point at the thing being demonstrated.

Decision, part 2 — placement: **this specification is extended in place**, not superseded by a feature 003. The new kind joins the existing passage registry, the existing two-valued precision model and the existing mention join. A second specification would have had to restate all three, and a restatement is a second source of truth — the same failure mode the reconciliation rules (contract §2.3) and the “one minter” rule (D-KG-2) are arranged against.

Decision, part 3 — sequencing: **specify now, build later**, after the 28 tasks this specification already carries unbuilt. Writing the tasks now fixes the accuracy obligation, the deduplication rule and the precision decision **before** anybody writes the pipeline that would otherwise settle them by accident.

Why this is the safe choice: the risk in OCR is not that it fails to run — it runs on this host. The risk is that it runs **well enough to look right**, deposits thousands of passages whose text is approximately correct and whose timings are approximately placed, and thereby degrades the accuracy of a corpus that currently has a measured one. Deciding scope without deciding the accuracy obligation would have realised

exactly that risk. So the obligation is decided here with the scope: per chapter, on two axes, against a budget derived from the recorded speech calibration rather than picked (FR-064, FR-064a, FR-064b; SC-031 to SC-034).

The consequence, recorded rather than discovered later: spec 002 cannot close until OCR accuracy is measured. This decision moved the finish line. It is written into the Clarifications block, into the task list's global constraints, and into Phase 11's checkpoint, so that a later reader arriving at "94 of 122 done" cannot mistake the remaining work for polish.

Honest boundary: three things about this decision are **not** claims of capability. (a) The OCR engine resolves on this host — measured 2026-09-02, tesseract at a user-local path carrying three language packs — but *no OCR artifact of any kind exists in the workshop today*, so every task below is genuinely unbuilt. (b) The new kind joins a precision model whose word branch is **contracted and currently unexercised**: workshop/docs/limits.md §10.1 and the defects row precision-segment-only record that every deep link in this deployment resolves at segment precision. (c) No number in this decision is an accuracy figure. None has been measured, and none may be written down until one is.

What would reopen it: a measured OCR error rate so far above the recorded speech-recognition baseline that on-screen mentions would degrade the corpus rather than extend it — in which case the honest response is to publish the figure and withhold the mentions, which FR-064b already requires, rather than to lower the budget until the figure passes.